Workflow-transfer benchmark of optical surface-roughness measurements against an internal tactile baseline across engineering materials and manufacturing processes

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Abstract

This study benchmarks exported optical roughness workflows against an internal tactile profilometry baseline across six engineering materials and multiple surface-generation processes. Rather than testing formal optical–tactile equivalence, the analysis examines which optical system–illumination workflows warrant prospective validation for specific material–process–parameter groups, and where tactile confirmation remains necessary. The main contribution is a reproducible workflow-level benchmark that integrates retrospective lowest-discrepancy, fixed-configuration, and leave-one-surface transfer analyses, thereby distinguishing local retrospective optimisation from more transferable performance. Eight profile roughness parameters were analysed. Optical–tactile discrepancy depended strongly on workflow choice. Among the lowest-discrepancy system–illumination combinations available in the archive, the smallest median discrepancies were observed for amplitude parameters including R_t, R_v, R_z, and R_{z1max}. Exported R_{sm} values required separate treatment because of scale sensitivity: multiplying retained optical R_{sm} values by 1000 reduced the retrospective lowest-discrepancy median from 99.9% to 17.3%, although 25% of material–process groups still exceeded 30%. Across strictly positive parameters, tactile random variability was generally smaller than the observed between-workflow discrepancies; the relative Type A standard uncertainty of the tactile mean had a median of 1.45%. A fixed-configuration sensitivity analysis showed that the complete-coverage fixed workflow with the lowest median discrepancy achieved 24.8%. In a leave-one-surface workflow-transfer check, selecting workflows from other material–process groups produced a held-out median discrepancy of 32.8%, with 52% of held-out groups still exceeding 30%. Overall, the resulting decision summaries establish a workflow-level benchmark for selecting optical candidates for prospective validation, while clearly distinguishing such candidates from claims of formal optical–tactile equivalence.

Keywords: surface metrology; optical measurement; tactile reference; surface roughness; illumination wavelength; benchmarking; uncertainty.

1 Introduction

Surface texture measurement remains central to manufacturing, where reported roughness parameters directly influence process control, specification compliance, and product traceability. In industrial practice, acceptance criteria are still commonly anchored to profile parameters such as R_a, R_q, R_{sk}, R_{sm}, R_t, R_v, R_z, and R_{z1max}, as defined in the ISO geometrical product specification framework for profile surface texture [1]. This study focuses on profile rather than areal parameters for three practical reasons. First, profile-based acceptance criteria remain dominant in the industrial sectors from which the archive was drawn—machining, finishing, and quality control—and the archived tactile reference workflow was a stylus profilometer, which is inherently a profile instrument. Second, the retained optical outputs include exported profile-parameter labels derived from profile-extraction procedures applied to areal data by the instrument software; the areal maps themselves were not archived consistently, precluding a harmonised areal-to-areal comparison. Third, profile parameters offer a common, standardised comparison language across the tactile and optical workflows represented in the archive. While areal parameters provide richer spatial information and are preferable for many modern surface-function analyses, the present benchmark is deliberately scoped to the exported profile-parameter label space that the archived instruments reported natively.

These profile parameters serve distinct industrial functions. R_a remains the most widely specified roughness parameter in machining drawings and quality-control standards, valued for its robustness and long regulatory history. R_z and R_{z1max} capture peak-to-valley extremes that are directly relevant to sealing, contact stiffness, and coating adhesion—applications where isolated deep valleys or high peaks dominate functional failure. R_{sm} describes the mean spacing of profile irregularities and is used in tribological and appearance-related specifications where texture periodicity matters. R_{sk} characterises the asymmetry of the height distribution and is important for bearing, lubrication, and running-in behaviour. R_t and R_v provide extreme-value descriptors used in fatigue and crack-initiation contexts. The benchmark therefore covers a parameter set that spans the main functional categories: amplitude (R_a, R_q, R_t, R_v, R_z, R_{z1max}), spacing (R_{sm}), and shape (R_{sk}).

Table 1 provides a structured summary of selected optical–tactile surface-texture comparison studies. Most previous work compared one or two instruments on a narrow material–parameter range, typically reporting best-case comparability rather than quantifying the effect of fixing or transferring the workflow. Interlaboratory comparisons [2] provide valuable reproducibility evidence but are restricted to calibrated standards and a single parameter. Application-focused studies on additively manufactured Ti6Al4V [3] illustrate the practical difficulty of measuring surface roughness when process-parameter variation strongly modulates the surface state, but they compare only profilometry against stereomicroscopy without a multi-workflow optical benchmark. The present study differs in combining six materials, four optical families, eight profile parameters, and three complementary analytical views (local selection, fixed-workflow sensitivity, held-out transfer) in a single, reproducible benchmark.

Table 1: Structured comparison of selected optical–tactile surface-texture studies. Abbreviations: CM = confocal microscopy; CSI = coherence scanning interferometry; FV = focus variation; WLI = white-light interferometry; FP = fringe projection; OCT = optical coherence tomography.

Study	Methods compared	Materials	Parameters	Key finding	Design
Mínguez-Martínez et al. (2022) [2]	Stylus vs. CM	Metallic roughness standard	R_a	Calibrated CM comparable with stylus for R_a on standards; ISO 17025 context	Interlaboratory comparison
Boedecker et al. (2010) [4]	WLI vs. tactile	Shape standards	Shape, R_a	Transfer-characteristic control and filtering required for comparability; spatial-frequency cut-off concept	Single-condition comparison
García et al. (2018) [5]	Stylus vs. CM	Not specified (methodological)	R_a, R_q, R_z	Stylus more reliable and faster; CM offers higher vertical and lateral resolution	Method comparison
Schmidt et al. (2021) [6]	Tactile vs. fringe projection	Machined components (steel)	Areal and profile parameters	Parameter validity shifts with application and material; areal/profile differences documented	Single-condition comparison
Nouira et al. (2014) [7]	Tactile vs. optical (confocal chromatic)	V-groove artefacts, VLSI standard, aspherical lens	Form, roughness	High-precision profilometer enables nanometre-level tactile–optical comparison of standards	Instrument development + comparison
Dierke & Tutsch (2012) [8]	Optical (layer-thickness)	Transparent materials	Layer thickness	Material influences on optical measurement; correction procedures	Method study
Leonhardt & Tiziani (1999) [9]	CSI, CM vs. tactile	Engineering surfaces	Microstructure	Confocal and WLI offer high-resolution measurements comparable with tactile; material aspects analysed via ellipsometry	Method comparison
Marichamy et al. (2025) [3]	Profilometry, stereomicroscopy	Ti6Al4V (L-PBF)	R_a (top surface, internal hole)	Process-parameter variation strongly modulates roughness; hole-quality measurement challenging for AM	Application study
This study	**CM, FV, fusion, CSI vs. tactile**	**6 materials, 12+ processes**	**8 profile parameters (R_a–R_{z1max})**	**Three-view benchmark: local selection, fixed-workflow, held-out transfer; selection penalty quantified**	**Multi-condition benchmark**

Recent work nevertheless illustrates substantial progress in low-force tactile/optical probing. Linz et al. presented 3D fibre probes combining tactile contact with optical read-out and very small contact force [10]. These developments expand the metrology toolbox while leaving a central comparability issue: optical and tactile systems may interrogate different effective measurands.

That comparability issue has been recognised repeatedly. Dierke and Tutsch addressed optical layer-thickness measurement of transparent materials [8]. Leonhardt and Tiziani addressed optical topometry of surfaces with locally changing materials, layers, and contaminations [9]. García et al. compared contact stylus and confocal microscopy methods from metrological and operational viewpoints [5]. Schmidt et al. compared areal and profile methods for machined components and discussed parameter validity for different applications and materials [6]. Nouira et al. described a high-precision profilometer for comparing tactile and optical probes against laser-interferometer measurements and standards [7]. Hoefling et al. addressed optical profilometry requirements for sheet-metal forming, including large object size, short measuring time, vertical resolution, and shining oil-covered metallic surfaces [11]. Stavroulakis et al. addressed external shape measurement using artificial intelligence and optimised data fusion for industrial applications [12]. The practical question concerns the material–process conditions under which a given optical workflow remains acceptably close to an established tactile baseline.

The physical origin of these discrepancies is also workflow-dependent. Optical roughness and topography values can be affected by the interaction between material or layer response, surface topography, spatial-frequency transfer, filtering, phase or envelope evaluation, lateral sampling, and profile-extraction choices [4,5,8,9,13]. Tactile values are instead influenced by stylus geometry, probing force, mechanical filtering by the tip, scan direction, contact accessibility, and profile specification operators [14–16]. Consequently, the same exported profile parameter may represent different effective measurands unless the acquisition and processing chain is controlled explicitly.

Workflow behaviour may also vary with illumination and reconstruction strategy. Hu et al. developed a line-scanning chromatic confocal sensor for 3D profile measurement on highly reflective materials [17]. Hinz et al. compared fringe-projection systems across scale ranges and emphasised application-specific scanner selection and data evaluation in production metrology [18]. Wang and Yang addressed saturation on highly reflective surfaces by combining adaptive and original-inverse fringe projection [19]. Schreiner et al. presented grazing-incidence infrared interferometry as a route to reduced speckle on optically rough technical surfaces [20]. Tiziani et al. discussed spectral and temporal phase evaluation for interferometry and speckle applications, including optically rough surfaces [13]. Sato and Ando recently proposed polynomial-exponent modelling of white-light scanning interferograms for envelope and fringe-phase identification [21].

Computational work places reconstruction and interpretation within the metrology chain. Xue et al. review machine-learning applications in optical metrology, including phase demodulation, unwrapping, phase-to-height conversion, intelligent sampling, noise reduction, and real-time processing [22]. Roberts et al. give industrial DUV–Vis–IR examples in which machine learning supplements traditional optical-metrology analysis models [23]. Chen et al. use spectrum-pattern matching to reconstruct chromatic confocal surface profiles [24]. Ekberg et al. use automated boundary detection to turn OCT data into dimensional measurements of multilayer ceramic structures [25]. Recent multi-task learning work also illustrates how shared models can be useful when multiple targets are learned from limited, sparse, or multi-instrument data [26, 27]. Neethu and Vinod Chandra review challenges caused by imbalanced and unlabeled data in machine learning [28], while Pouchard et al. document reproducibility problems caused by training-data issues, platform/version differences, randomness, and hyperparameter variation in materials-informatics machine learning [29]. Heterogeneous retrospective datasets also constrain the level of inference supported by such approaches.

Existing optical–tactile comparisons often report comparability for selected instruments, surfaces, or parameters, but they rarely separate three operational questions: whether a low-discrepancy optical candidate exists locally, whether one workflow can be fixed before measurement across a broad surface range, and whether workflow choices transfer to unseen material–process classes. The present archive enables this separation and is therefore used as a decision-oriented benchmark rather than as a direct equivalence study.

The contribution is a structured, reproducible benchmark of workflow discrepancy within the analysed archive. Its novelty is the combination of three complementary views of optical workflow performance: local lowest-discrepancy selection, fixed-configuration sensitivity, and held-out surface-class transfer. This design makes the optimisation effect visible instead of reporting only the most favourable optical–tactile matches. The comparison space is defined by matched exported parameter labels and is used to characterise workflow-level discrepancy patterns for follow-up validation.

2 Methods

2.1 Study design and unit of analysis

This work presents a retrospective benchmark based on an archived dataset of tactile and optical measurements acquired across six engineering materials and multiple surface-generation routes. Each observation is defined by specimen identity, roughness parameter, optical system, illumination condition, optical output, and the corresponding tactile result. The primary unit of comparison is the parameter–surface group, defined as a single roughness parameter within a given material–process combination. Within the manuscript subset, each of the eight parameters contributes 59 material–process groups, giving a total of 472 parameter–surface groups. Because not every optical workflow is represented in every group, all analyses were conducted on the available comparable subset for each task.

Archive specimen labels follow the pattern `<material>_<process>` (for example, `ti6al4v_mf` or `14301_wedm_f`). Material identifiers denote C45 steel (`c45`), AISI 304 stainless steel / EN 1.4301 (`14301`), Ti6Al4V (`ti6al4v`), Al7075 (`al7075`), MO58A brass (`mo58a`), and ELLOR graphite (`ellor`). Process identifiers denote rough milling (`mr`), finish milling (`mf`), rough and finish face turning (`tr`, `tf`), burnishing after finish milling (`b`), rough and finish wire electrical discharge machining (`wedm_r`, `wedm_f`), grinding (`gri`), glass bead blasting (`gla`), and honing (`hon`). The manuscript tables expand these archive codes for readability, whereas the exported archive files retain the short slugs.

The process labels denote archived surface-generation route classes rather than complete manufacturing recipes. Detailed process metadata—including tool geometry, feed and speed settings, depths of cut, wire EDM discharge settings, and abrasive or honing conditions—were retained unevenly across materials. Accordingly, material–process groups were analysed as benchmark strata rather than as factors in a designed manufacturing experiment.

2.2 Tactile baseline

Tactile stylus profilometry was treated as the internal baseline because it was the only workflow in the archive for which repeated measurements were retained consistently across the manuscript subset. The terminology for measurement uncertainty follows VIM and GUM usage, distinguishing repeatability descriptors from a traceability-qualified uncertainty budget [30,31]. For each specimen and roughness parameter, the tactile mean

$$S = \frac{1}{n_S}\sum_{k=1}^{n_S} T_k$$

was computed from $n_S = 20$ repeats, with accompanying repeatability standard deviation s_S and Type A standard uncertainty of the mean $u_S = s_S/\sqrt{n_S}$. These quantities describe internal repeatability and the Type A component available from the archive; they are reported separately from traceability-qualified uncertainty.

The tactile mean was therefore used as an internal baseline rather than as a fully traceability-qualified reference value. Although the retained archive supports Type A repeatability descriptors for tactile measurements, it does not provide a complete combined uncertainty budget incorporating stylus geometry, probing force, calibration artefact uncertainty, filtering choices, optical vertical and lateral calibration, optical repeatability, and reconstruction settings. Prospective validation would additionally require documented contact-instrument characteristics and profile specification operators, including stylus geometry, nominal contact conditions, evaluation length, and profile assessment rules consistent with profile GPS standards [14,15]. Consequently, the reported values quantify exported-workflow discrepancy within the archive and should be interpreted as screening evidence for prospective validation.

2.3 Optical workflows and comparison space

The optical archive includes confocal microscopy, focus variation, confocal–interferometric fusion, and interferometry-based systems, with multiple illumination bands where available. System abbreviations used in the figures and tables follow the archive nomenclature: *conf* for confocal microscopy, *FV* for focus variation, *fusion* for confocal–interferometric fusion, and *int* for interferometry/coherence-scanning interferometry.

Comparison was restricted to matched specimen identity and same-named exported roughness parameters. The retained optical metadata identify workflow family and illumination condition; retained row labels also encode nominal x10 and 5x5 acquisition labels, but not enough numerical information to reconstruct objective, sampling, filtering, form-removal, or reconstruction settings across all instruments. The retained values are instrument-exported profile-parameter labels rather than raw areal maps or reconstructed profile traces, so profile direction, profile averaging, lateral resampling, and profile-extraction settings cannot be harmonised uniformly. A prospective raw-profile comparison would require a common filtering and form-removal chain, including ISO/GPS profile-filter choices [16]. The reported discrepancies are between-workflow discrepancies within this exported-parameter comparison space.

Table 2: Retained archive fields and metrological role in the workflow benchmark

Information element	Retention in archive	Role in the present analysis
Material, process, and specimen identifiers	Retained	Define material–process strata and matched comparison groups
Profile roughness parameter labels	Retained	Define same-named exported-parameter comparisons
Tactile repeats	Retained consistently ($n_S = 20$)	Internal baseline mean and repeatability descriptor
Optical workflow family and illumination condition	Retained	System+illumination ranking, fixed-workflow analysis, transfer check
Optical exported roughness output	Retained	Optical–tactile discrepancy calculation
Nominal acquisition labels	Partially retained	Qualitative archive metadata; not sufficient for full reconstruction
Raw tactile/optical profiles and areal maps	Not retained uniformly	Excluded from common raw-profile measurand reconstruction
Filtering, form removal, and profile-extraction settings	Not retained uniformly	Treated as part of the workflow-level comparison frame
Optical objective, sampling, resolution, and calibration metadata	Not retained uniformly	Discussed as prospective-validation requirements
Combined tactile–optical uncertainty budget	Not retained	Outside the retrospective benchmark scope

2.4 Response variables

The parameter symbols follow the profile roughness notation used in ISO surface-texture terminology. The comparison is restricted to same-named exported profile parameters; areal surface-texture parameters were outside the retained manuscript subset. For the seven strictly positive roughness parameters (R_a, R_q, R_{sm}, R_t, R_v, R_z, and R_{z1max}), discrepancy between optical output O and tactile baseline S was defined as

$$d_p = 100 \left| \frac{O - S}{S} \right|. \tag{1}$$

For the sign-sensitive parameter R_{sk}, near-zero tactile denominators make percentage normalisation unstable. Archive-derived decision or regression summaries that retain percentage-based R_{sk} values are treated as secondary screening descriptors, separately from the strictly positive parameters. The per-group percentage-based R_{sk} screening summaries are retained in the supplementary CSV output `results_decision_rsk_screening_groups.csv` for archive completeness. No alternative R_{sk} normalisation (absolute difference or sign-consistency index) is computed in the current analysis package; these remain available for future extension.

2.5 Decision-oriented summaries

Within each parameter–surface group, optical workflows were ranked by the available observed discrepancy, using the median when more than one retained optical output was available for the same system–illumination pair. In the archived decision subset, optical replication is incomplete; most system–illumination entries represent one exported optical result matched to a tactile mean. The decision tables report the minimum retained discrepancy across the tested system–illumination combinations and identify the workflow that achieved it. These retrospective summaries describe lowest-discrepancy behaviour within the archive, separately from the performance expected for a single fixed optical configuration.

To quantify this optimisation effect, a fixed-configuration sensitivity analysis was performed for the seven strictly positive parameters. Each candidate optical workflow was defined as one system–illumination pair and was evaluated across all parameter–surface groups in which that pair was available, without allowing the workflow to vary by group. The resulting fixed-workflow medians were compared with the retrospective lowest-discrepancy medians from the same groups. Coverage was reported relative to the 413 strictly positive parameter–surface groups used in the decision summaries.

A leave-one-surface workflow-transfer check was then performed to estimate how well workflow choices selected from the archive transfer to a held-out surface class. For each strictly positive parameter and held-out material–process group, system–illumination workflows were ranked using all other material–process groups for the same parameter. The highest-ranked workflow available for the held-out group was then evaluated on that held-out group. This analysis remains retrospective and archive-based, but the held-out group is not used to rank candidate workflows.

The retained metadata are summarised in Table 2. Accordingly, the release supports a retrospective workflow benchmark rather than a prospective traceability-oriented validation study.

The 10% and 30% discrepancy bands used in the decision summaries are operational triage bands. They are not tolerance limits, conformance criteria, or substitutes for application-specific measurement uncertainty statements.

2.6 Statistical analysis

Statistical analyses were performed within parameter–surface groups. Two-factor ANOVA with factors *system* and *illumination colour* provided an exploratory decomposition of variance. In this retrospective, unbalanced archive, nominal p-values served as screening statistics; no family-wise confirmatory inference was claimed across the large number of per-group screens. Interpretation was based primarily on effect sizes and on consistency of patterns across groups.

System-wise linear regressions of discrepancy versus nominal illumination wavelength were used as low-degree-of-freedom descriptive summaries. Most fits rely on three or four wavelength levels and were treated as screening trends rather than physical calibration laws.

Non-parametric percentile bootstrap intervals were computed for the aggregate retrospective lowest-discrepancy medians by parameter, material, and process. Bootstrap resampling was performed across archive-defined parameter–surface groups within each aggregate stratum, using 2000 resamples and a fixed random seed; these intervals describe stability of the retrospective median summaries and are not prospective validation intervals.

Supplementary tables and representative figures were generated from the same exported analysis outputs used in the main manuscript. The supplement generator (`generate_supplement.py`) formats the ANOVA screens, wavelength-regression summaries, system descriptives, tactile repeatability summaries, bootstrap stability summaries, and selected representative figures without adding a separate statistical model.

To compare workflow similarity within a given parameter–surface group, a technical distance matrix was defined from mean discrepancies. If $\bar{d}_i$ and $\bar{d}_j$ denote mean discrepancy for techniques i and j, where a technique is one optical system under one illumination condition, then

$$D_{ij} = |\bar{d}_i - \bar{d}_j|. \tag{2}$$

This quantity describes similarity of discrepancy patterns within the analysed dataset, separately from traceability, matched calibration, or measurand equivalence.

3 Results

Across the manuscript subset, each of the eight parameters contributes 59 material–process groups, yielding 472 parameter–surface groups overall, all with tactile baselines based on $n_S = 20$ repeats. For the seven strictly positive parameters, the relative Type A standard uncertainty of the tactile mean had median 1.45% and interquartile range 0.85–2.44%; parameter-wise summaries are reported in Supplementary Table S4. The random variability of tactile measurements is generally smaller than the between-workflow discrepancies analysed below, supporting the use of the tactile mean as an internal benchmark baseline.

3.1 Decision-oriented benchmark summaries

For each parameter–surface group, the decision summaries retain the lowest observed median discrepancy across the tested system and illumination combinations. These tables describe retrospective lowest-discrepancy performance within the archive, separately from the behaviour of any single fixed configuration.

Table 3 summarises retrospective lowest-discrepancy medians for the seven strictly positive parameters. Among them, amplitude-type measures (R_t, R_v, R_z, and R_{z1max}) achieve the lowest medians, ranging from 4.8% to 6.6%; R_a and R_q remain below 10% in median terms, whereas R_{sm} remains problematic at 99.9%. No single optical workflow dominates all parameters, although interferometry-based configurations are frequently among the lowest-discrepancy options. Percentage-based R_{sk} rankings are excluded from the manuscript decision tables due to near-zero tactile denominator instability; they are retained as archive-level screening outputs.

The band columns in Table 3 provide a practical triage view rather than a normative acceptance rule. Groups in the lowest band identify candidate optical workflows for prospective protocol validation, groups in the middle band indicate cases where material–process-specific checking remains important, and groups above 30% describe conditions where tactile confirmation or a revised optical protocol remains the more defensible route. The same interpretation separates R_{sm} from the amplitude parameters: all 59 R_{sm} groups remain in the high-discrepancy band, whereas the lower-discrepancy band contains 63–69% of the R_t, R_v, R_z, and R_{z1max} groups.

Table 3: Decision-oriented summary of optical–tactile discrepancy by strictly positive roughness parameter (lowest-discrepancy system+colour chosen per material–process group)

Parameter	N	Lowest median |diff| [%]	Q1–Q3	$\leq$ 10%	10–30%	>30%	Most frequent lowest (system, colour)
R_a	59	9.8	2.6–21.8	30 (51%)	16 (27%)	13 (22%)	int, red
R_q	59	7.4	2.4–22.5	35 (59%)	12 (20%)	12 (20%)	int, red
R_{sm}	59	99.9	99.9–99.9	0 (0%)	0 (0%)	59 (100%)	int, white
R_t	59	6.6	2.1–13.2	39 (66%)	13 (22%)	7 (12%)	int, green
R_v	59	4.8	1.3–19.0	40 (68%)	7 (12%)	12 (20%)	int, red
R_z	59	5.3	1.2–20.3	37 (63%)	10 (17%)	12 (20%)	int, red
R_{z1max}	59	5.7	2.0–12.3	41 (69%)	10 (17%)	8 (14%)	int, red

Note: N is the number of available material–process groups for the seven strictly positive parameters. For each group, the optical configuration (system+illumination colour) that minimises the median $|\text{diff}|(\%)$ is selected. Q1–Q3 are quartiles of the selected medians. The band columns ($\leq 10\%$, 10–30%, >30%) count how many groups fall into each range; values are shown as count (percent of N), e.g. 7 (15%) means 7 groups, i.e. 15% of N. Most frequent lowest denotes the modal selected (system, colour) across groups. Percentage-based R_{sk} rankings are excluded from the manuscript decision tables because normalisation becomes unstable near zero; archive-level screening rows are retained separately in the exported outputs.

Because R_{sm} was the only strictly positive parameter that remained uniformly in the high-discrepancy band, it was segmented separately by optical system, material, and process (Table 4). The diagnostic segmentation shows that the effect is not concentrated in one workflow family or one surface class. For all four optical-system families, every retained R_{sm} system–colour entry exceeded 30% discrepancy. After retrospective lowest-discrepancy workflow selection, every material stratum and every process stratum also remained entirely above 30% discrepancy.

Table 4: Diagnostic segmentation of R_{sm} optical–tactile discrepancy by optical system, material, and process

View	Stratum	N	Median |diff| [%]	Q1–Q3	$\leq 10\%$	>30%	Modal workflow
Optical system	conf	172	100.0	99.9–100.0	0 (0%)	172 (100%)	all colours
Optical system	FV	177	99.9	99.9–99.9	0 (0%)	177 (100%)	all colours
Optical system	fusion	177	100.0	100.0–100.0	0 (0%)	177 (100%)	all colours
Optical system	int	233	99.9	99.9–99.9	0 (0%)	233 (100%)	all colours
Material	AISI 304 / EN 1.4301 (14301)	10	99.9	99.9–99.9	0 (0%)	10 (100%)	FV, white
Material	Al7075	10	99.9	99.8–99.9	0 (0%)	10 (100%)	FV, white
Material	C45 steel	10	99.9	99.9–99.9	0 (0%)	10 (100%)	int, white
Material	Graphite (ELLOR)	9	99.9	99.9–99.9	0 (0%)	9 (100%)	int, blue
Material	MO58A brass	10	99.9	99.9–99.9	0 (0%)	10 (100%)	FV, green
Material	Ti6Al4V	10	99.9	99.9–99.9	0 (0%)	10 (100%)	int, white
Process	MR (rough milling)	6	99.9	99.8–99.9	0 (0%)	6 (100%)	int, white
Process	MF (finish milling)	6	99.9	99.9–100.0	0 (0%)	6 (100%)	int, white
Process	TR (rough face turning)	6	99.8	99.7–99.8	0 (0%)	6 (100%)	int, white
Process	TF (finish face turning)	6	99.8	99.6–99.9	0 (0%)	6 (100%)	int, white
Process	B (burnishing after MF)	5	99.9	99.9–100.0	0 (0%)	5 (100%)	int, white
Process	WEDM_R (rough wire EDM)	6	99.9	99.9–99.9	0 (0%)	6 (100%)	int, blue
Process	WEDM_F (finish wire EDM)	6	99.9	99.9–99.9	0 (0%)	6 (100%)	FV, green
Process	GRI (grinding)	6	99.9	99.9–99.9	0 (0%)	6 (100%)	FV, green
Process	GLA (glass bead blasting)	6	99.9	99.9–99.9	0 (0%)	6 (100%)	FV, green
Process	HON (honing)	6	99.9	99.9–99.9	0 (0%)	6 (100%)	FV, green

Note: Optical-system rows summarise all retained R_{sm} system–colour entries. Material and process rows summarise the retrospective lowest-discrepancy R_{sm} result per material–process group. The modal workflow is the most frequently selected lowest-discrepancy system+colour within the corresponding material or process stratum; for optical-system rows, all retained illumination colours are pooled.

The retained R_{sm} comparison was then examined for unit-scale sensitivity because optical spacing exports were numerically much smaller than the tactile spacing values. Multiplying optical R_{sm} values by 1000, treated as a post hoc millimetre-to-micrometre sensitivity rather than as a primary correction, reduced the all-entry median discrepancy from 99.9% to 47.4% and the lowest-discrepancy workflow median across material–process groups from 99.9% to 17.3% (Table 5). Under this sensitivity, 24 of 59 lowest-discrepancy workflow groups moved into the $\leq 10\%$ band, while 15 of 59 remained above 30%. Accordingly, retained exported-value R_{sm} is not used to support practical optical-substitution recommendations; it is retained as a harmonisation-sensitive diagnostic pending raw-profile validation.

Table 5: Sensitivity of R_{sm} discrepancy to retained optical unit scale

Comparison set	Scale assumption	N	Median |diff| [%]	Q1–Q3	$\leq 10\%$	>30%	Median tactile/optical ratio
All retained system-colour entries	Exported optical value	761	99.9	99.9–100.0	0 (0%)	761 (100%)	1799.8
All retained system-colour entries	Optical value multiplied by 1000	761	47.4	28.0–66.0	53 (7%)	558 (73%)	1.8
Lowest-discrepancy workflow per material-process group	Exported optical value	59	99.9	99.9–99.9	0 (0%)	59 (100%)	1030.4
Lowest-discrepancy workflow per material-process group	Optical value multiplied by 1000	59	17.3	3.9–30.2	24 (41%)	15 (25%)	1.2

Note: The x1000 rows are a sensitivity check that treats retained optical R_{sm} values as if they were exported in millimetres and converted to micrometres before comparison. They are not used as corrected primary endpoints because complete original unit metadata and raw processing chains are not retained consistently across the archive.

Tables 6 and 7 aggregate the same retrospective lowest-discrepancy summaries by surface-generation process and by material across the 413 strictly positive parameter–surface groups. Even after allowing the system and illumination to vary, process and material still strongly modulate how often optical workflows enter a low-discrepancy regime.

Bootstrap percentile intervals for the aggregate medians are reported in Supplementary Table S5. They support the same hierarchy as the point estimates: amplitude-parameter medians remain in the lower-discrepancy range, retained exported-value R_{sm} remains close to 100%, and process-level intervals are wider where the archive combines more heterogeneous surface-generation routes.

Table 6: Decision-oriented discrepancy summary aggregated by surface-generation process across all materials and strictly positive parameters

Process	N	Lowest median |diff| [%]	Q1–Q3	$\leq 10\%$	10–30%	>30%	Most frequent lowest (system, colour)
GLA (glass bead blasting)	42	4.7	1.1–7.5	35 (83%)	1 (2%)	6 (14%)	FV, blue
WEDM_R (rough wire EDM)	42	1.6	0.6–8.1	33 (79%)	3 (7%)	6 (14%)	int, green
HON (honing)	42	3.8	1.2–12.1	30 (71%)	5 (12%)	7 (17%)	int, red
WEDM_F (finish wire EDM)	42	7.5	2.9–12.9	29 (69%)	7 (17%)	6 (14%)	FV, green
MF (finish milling)	42	8.1	2.0–24.0	23 (55%)	11 (26%)	8 (19%)	int, green
GRI (grinding)	42	11.2	2.8–94.7	18 (43%)	8 (19%)	16 (38%)	int, blue
MR (rough milling)	42	28.6	4.7–42.7	17 (40%)	4 (10%)	21 (50%)	int, blue
TF (finish face turning)	42	25.3	3.0–50.2	16 (38%)	5 (12%)	21 (50%)	fusion, red
TR (rough face turning)	42	31.2	7.6–75.0	12 (29%)	9 (21%)	21 (50%)	fusion, red
B (burnishing after MF)	35	19.0	9.7–37.7	9 (26%)	15 (43%)	11 (31%)	int, white

Note: N is the number of contributing strictly positive parameter–material–process groups. For each group, the optical configuration (system+illumination colour) that minimises the median $|\text{diff}|(\%)$ is selected. Q1–Q3 are quartiles of the selected medians. The band columns ($\leq 10\%$, 10–30%, >30%) count how many groups fall into each range; values are shown as count (percent of N), e.g. 7 (15%) means 7 groups, i.e. 15% of N. Most frequent lowest denotes the modal selected (system, colour) across groups. Archive identifiers in the first column are expanded for readability.

Table 7: Decision-oriented discrepancy summary aggregated by material across all processes and strictly positive parameters

Material	N	Lowest median $\lvert$diff$\rvert$ [%]	Q1–Q3	$\leq 10\%$	10–30%	>30%	Most frequent lowest (system, colour)
Graphite (ELLOR)	63	4.6	0.9–19.8	42 (67%)	6 (10%)	15 (24%)	int, green
C45 steel	70	7.3	1.2–54.6	40 (57%)	6 (9%)	24 (34%)	int, red
MO58A brass	70	8.2	3.4–33.1	38 (54%)	13 (19%)	19 (27%)	int, blue
Al7075	70	8.2	3.0–41.0	37 (53%)	11 (16%)	22 (31%)	int, red
AISI 304 / EN 1.4301 (14301)	70	10.9	1.6–33.3	33 (47%)	17 (24%)	20 (29%)	FV, green
Ti6Al4V	70	11.7	4.4–51.6	32 (46%)	15 (21%)	23 (33%)	int, red

Note: N is the number of contributing strictly positive parameter–material–process groups. For each group, the optical configuration (system+illumination colour) that minimises the median $\lvert$diff$\rvert(\%)$ is selected. Q1–Q3 are quartiles of the selected medians. The band columns ($\leq 10\%$, 10–30%, >30%) count how many groups fall into each range; values are shown as count (percent of N), e.g. 7 (15%) means 7 groups, i.e. 15% of N. Most frequent lowest denotes the modal selected (system, colour) across groups. Archive identifiers in the first column are expanded for readability.

The fixed-configuration sensitivity analysis makes the retrospective-selection effect explicit (Table 8). The lowest-median single workflow with complete coverage was interferometry with green illumination, with median discrepancy 24.8% across all 413 strictly positive parameter–surface groups and $\leq 10\%$ discrepancy in 93 groups (23%). Interferometry with white illumination achieved a similar median discrepancy (25.0%) but with slightly lower coverage (406/413 groups). These fixed-workflow medians are substantially higher than the retrospective lowest-discrepancy medians in Table 3, showing that the low discrepancies in the decision summaries depend on parameter- and surface-specific workflow selection.

Table 8: Fixed-configuration sensitivity analysis for the strictly positive roughness parameters

Fixed workflow	N	Coverage	Median |diff| [%]	Q1–Q3	$\leq 10\%$	>30%	Median excess vs. lowest [%]
int, green	413	100%	24.8	11.1–66.9	93 (23%)	198 (48%)	7.2
int, white	406	98%	25.0	10.8–66.7	92 (23%)	189 (47%)	8.0
int, red	413	100%	31.5	12.1–90.5	87 (21%)	212 (51%)	9.1
FV, green	413	100%	35.4	14.9–86.7	67 (16%)	232 (56%)	13.7
FV, blue	413	100%	40.2	15.5–99.9	66 (16%)	235 (57%)	11.5
int, blue	399	97%	43.0	19.9–90.3	48 (12%)	249 (62%)	13.5
conf, blue	378	92%	46.5	18.2–99.9	62 (16%)	236 (62%)	17.8
FV, red	413	100%	47.2	19.0–99.7	69 (17%)	265 (64%)	17.5

Note: Fixed workflow means that one optical system+illumination pair is evaluated wherever it is available, without selecting a different configuration for each parameter–surface group. Coverage is relative to the 413 strictly positive parameter–surface groups used in the manuscript decision summaries. The table shows the eight lowest median fixed workflows; the complete fixed-workflow ranking and group-level medians are exported as CSV files. Excess vs. lowest is the median difference between the fixed-workflow median discrepancy and the retrospective lowest-discrepancy median within the same group.

The leave-one-surface workflow-transfer check further reduces the influence of direct per-group optimisation (Table 9). When system–illumination rankings were learned from other material–process groups of the same parameter, the overall held-out median discrepancy was 32.8% (Q1–Q3 13.1–70.6%). Only 82 of 413 held-out groups (20%) reached the $\leq 10\%$ band, while 214 groups (52%) remained above 30%. The median excess over the retrospective lowest-discrepancy result in the same groups was 9.3 percentage points. The transfer penalty was especially visible for R_t, R_v, R_z, and R_{z1max}, where the retrospective lowest-discrepancy medians were low but held-out workflow choices often moved back into the middle or high-discrepancy bands.

Table 9: Leave-one-surface workflow-transfer check for the strictly positive roughness parameters

Parameter	N	Held-out median $\|\text{diff}\|$ [%]	Q1–Q3	$\leq 10\%$	>30%	Median excess vs. lowest [%]	Most frequent selected workflow
All	413	32.8	13.1–70.6	82 (20%)	214 (52%)	9.3	int, green
R_a	59	21.1	11.4–49.1	12 (20%)	20 (34%)	8.6	int, white
R_q	59	19.6	10.8–65.4	13 (22%)	24 (41%)	9.9	int, red
R_{sm}	59	99.9	99.9–100.0	0 (0%)	59 (100%)	0.0	FV, blue
R_t	59	37.0	16.3–58.6	9 (15%)	35 (59%)	21.9	int, green
R_v	59	27.7	7.6–54.3	17 (29%)	28 (47%)	12.3	int, green
R_z	59	20.4	7.2–54.0	19 (32%)	23 (39%)	9.0	int, green
R_{z1max}	59	20.8	11.4–47.3	12 (20%)	25 (42%)	13.4	int, white

Note: For each held-out material–process group, system+illumination workflows were ranked using all other material–process groups for the same parameter. The table evaluates the highest-ranked workflow that was available for the held-out group. The held-out group was not used to rank workflows. Excess vs. lowest is the median increase relative to the retrospective lowest-discrepancy result in the same held-out groups. This check estimates transfer of workflow selection across surfaces within the archive and is not a prospective validation interval.

Read together, Tables 3, 8, and 9 separate three operational decisions. The retrospective lowest-discrepancy summaries describe where the archive contains an optical workflow worth validating for a specific parameter–surface group. The fixed-workflow summaries describe the consequence of choosing one optical setup before measurement across a broad material–process range. The transfer check describes what happens when the workflow choice is learned from other surface classes rather than selected directly within the target group. This distinction is central to the benchmark: the archive contains many lower-discrepancy local matches, but the lowest-median complete-coverage fixed workflow still leaves nearly half of the strictly positive parameter–surface groups above 30% discrepancy, and learned workflow transfer leaves slightly more than half above that band.

Figure 1 summarises this selection penalty directly. Across the strictly positive parameters, retrospective within-group selection gives the lowest aggregate median discrepancy, the lowest-median complete-coverage fixed workflow increases the median discrepancy, and leave-one-surface transfer increases it further. The figure is intended as a compact visual check that the benchmark does not rely only on retrospectively favourable matches.

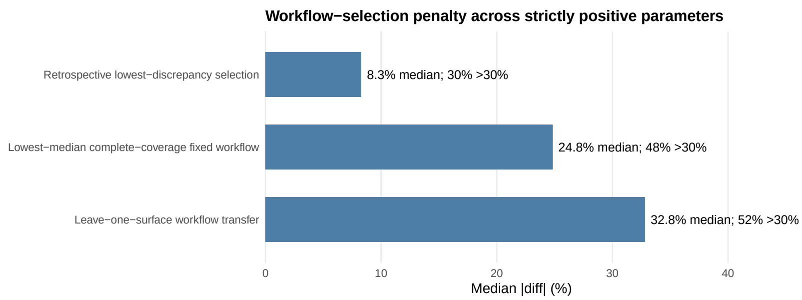

Figure 1: Workflow-selection penalty across the seven strictly positive roughness parameters. Bars show the median $|\mathrm{diff}|(\%)$ for retrospective lowest-discrepancy selection, the lowest-median complete-coverage fixed workflow, and leave-one-surface workflow transfer; labels also report the percentage of groups above 30% discrepancy.

Figure 2 summarises the same retrospective lowest-discrepancy benchmark as a process–parameter map. It is derived from the same retrospective workflow-selection logic as Table 3, but retains the process dimension. It highlights two practically important patterns. First, the retained exported-value R_{sm} comparison remains close to 100% median discrepancy across all processes even after retrospective workflow selection. Second, amplitude parameters show strong process dependence: rough wire EDM, honing, glass bead blasting, and finish wire EDM generally occupy the low-discrepancy region, whereas rough face turning and rough milling contain several higher-discrepancy cells.

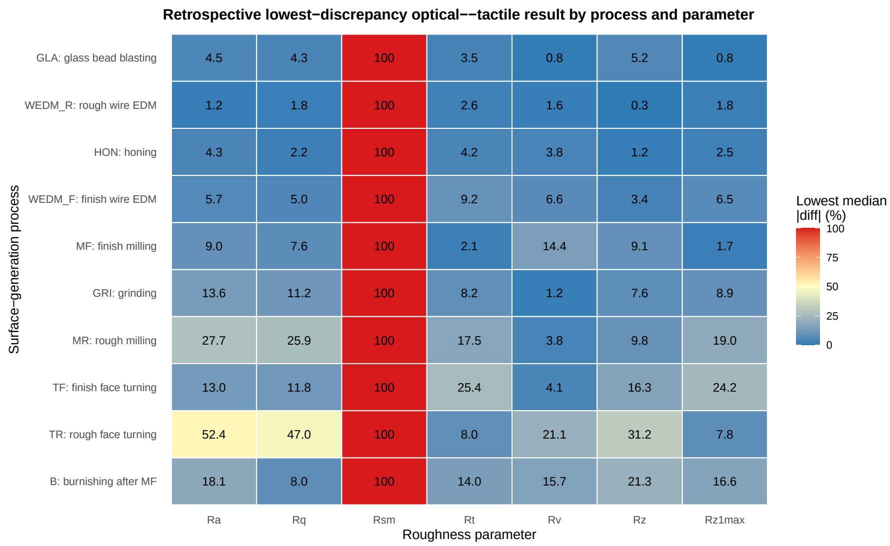

Figure 2: Global process–parameter map of retrospective lowest-discrepancy optical–tactile median discrepancy for the seven strictly positive roughness parameters. Values are median $|\text{diff}|(\%)$ after selecting the lowest-discrepancy available system–illumination pair within each material–process–parameter group; colour scale is capped at 100% for readability.

Across the full dataset, interferometry-based measurements (*int*) appear most frequently as the retrospective lowest-discrepancy configuration for most parameters, and red/green illumination bands are most often associated with the lowest-discrepancy median $|\text{diff}|(\%)$ (Table 3). Process- and material-specific exceptions are also present, with other systems, notably focus variation (FV), becoming the most frequent lowest-discrepancy choice in selected strata (Tables 6–7). These frequencies describe patterns within the tested archive.

Figure 3 illustrates the main effects of the measurement system on the absolute percentage error using the representative case of R_a on Ti6Al4V after finish milling. Analogous plots for other material–process combinations are included in the supplementary materials.

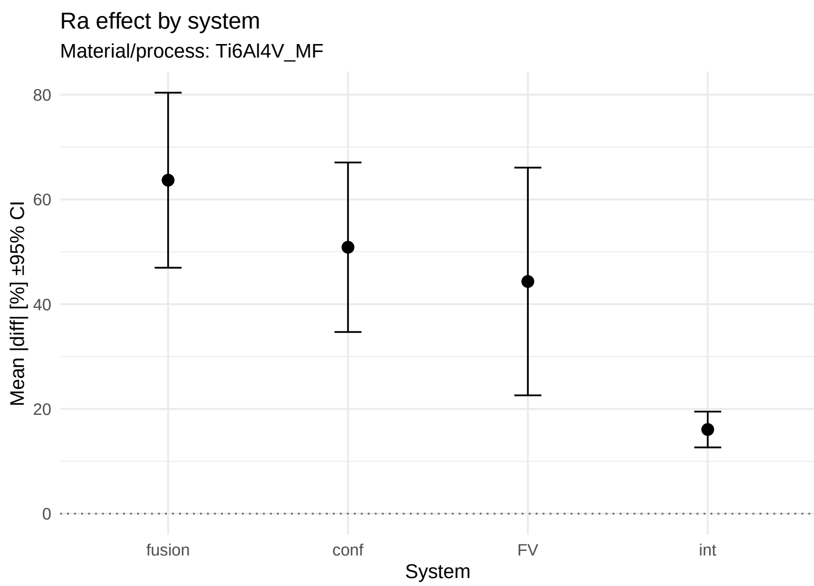

Figure 3: Main effects of the measurement system on absolute roughness error $|\mathrm{diff}|(\%)$ for R_a on Ti6Al4V after finish milling

3.2 Representative example: Ti6Al4V after finish milling

To illustrate the system-to-system variability observed across the full dataset, a detailed example is shown for Ti6Al4V after finish milling, which represents a technologically relevant, moderately smooth surface. In the decision summaries, roughly half of all available R_a material–process groups reach $\leq 10\%$ discrepancy after selecting the lowest-discrepancy optical system and colour (Table 3), while Ti6Al4V is among the more challenging materials in aggregate (Table 7).

For Ti6Al4V after finish milling, the spread of $|\mathrm{diff}|(\%)$ between systems is particularly clear. Some optical instruments yield errors close to the tactile reference under suitable illumination, whereas other configurations deviate markedly. Optical–tactile discrepancy is reported at the *system+illumination* level.

Figure 4 compares the distributions of $|\mathrm{diff}|(\%)$ for all optical systems against the tactile reference for Ti6Al4V after finish milling, emphasising the contrast between well-performing and outlying configurations.

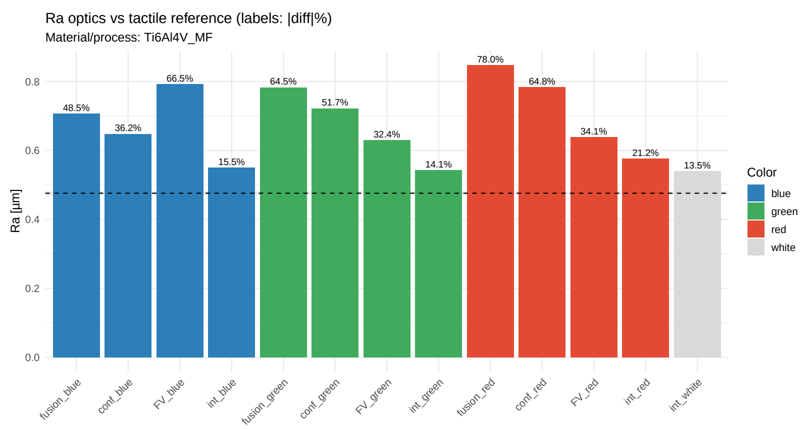

Figure 4: Comparison of absolute roughness error $|\text{diff}|(\%)$ for R_a on Ti6Al4V after finish milling obtained from optical systems and the tactile reference

3.3 Worst-case illustration: 14301 stainless steel after rough face turning

To balance the presentation, a representative worst-case example is provided for 14301 stainless steel after rough face turning. This material–process combination is consistently among the most challenging in the archive. For R_a, all optical workflows underestimate the roughness relative to the tactile baseline, with discrepancies ranging from approximately 42% to 60%. No system–illumination combination enters the $\leq 10\%$ band, and no configuration falls below 30%. The high discrepancies are attributed to the combined effect of the high reflectance of stainless steel in the visible range, the steep, directional tool marks produced by rough turning, and the resulting speckle and dynamic-range limitations in several optical systems. The contrast with the Ti6Al4V finish-milling case (Section 3.2) illustrates the material–process modulation range that underlies the aggregate summaries in Tables 7–6. The full per-group data for this worst-case example are available in the supplementary CSV outputs.

3.4 Influence of illumination wavelength

Across materials and processes, illumination colour can shift $|\text{diff}|(\%)$ and, in a minority of cases, induces systematic trends with wavelength. In the full set of system-wise regressions across the analysed material–process–parameter groups, statistically significant slopes β_1 of $|\text{diff}|(\%)$ versus wavelength were observed in 8.9% of fits (203/2290), indicating wavelength optimisation as an application-specific tuning step rather than a universal correction. The regressions span many parameter–surface groups and typically rely on only three or four colour levels per fit; individual significant slopes serve as screening evidence. The observed rate is only modestly above the nominal 5% false-positive rate, so the archive supports selective wavelength effects rather than a broad wavelength law.

Consistent with this, ANOVA effect sizes indicate that the measurement system accounts for substantially more variability in $|\text{diff}|(\%)$ than illumination colour for most groups (median $\eta^2_{\text{system}} = 0.651$ vs. median $\eta^2_{\text{color}} = 0.106$ across all ANOVA groups, $n = 590$; Supplementary Table S1). Across the same 590 parameter–surface groups, the system effect size exceeds the colour effect size in 497 cases, whereas colour exceeds system in 93. Nevertheless, colour still matters operationally because the lowest-discrepancy configuration for many parameters is tied to a specific illumination band (Table 3), and these preferences can differ across process and material (Tables 6–7).

The full per-group ANOVA outputs (including degrees of freedom, F, p, and η^2) are reported in Supplementary Table S1, and the full set of system-wise wavelength-regression slopes is reported in Supplementary Table S2. Correlation coefficients were used as a complementary descriptive diagnostic alongside regression; to avoid redundancy, the paper reports regression slopes as the primary summary. Additional system- and colour-wise effect plots supporting follow-up comparisons are provided in the supplementary materials.

Figure 5 shows an example of the dependence of $|\text{diff}|(\%)$ on illumination wavelength for R_a on Ti6Al4V after finish milling, together with fitted screening lines that illustrate the estimated trend for each optical system.

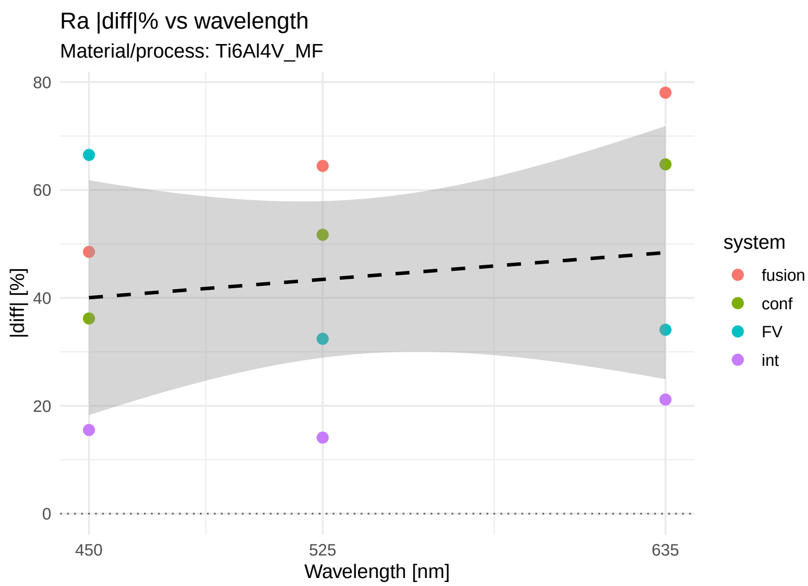

Figure 5: Dependence of absolute roughness error $|\text{diff}|(\%)$ for R_a on Ti6Al4V after finish milling on illumination wavelength for individual optical systems; lines indicate fitted screening trends

3.5 Technical distance between systems

The technical distance matrix derived from mean $|\text{diff}|(\%)$ highlights clusters of optical systems that behave similarly across the investigated materials and surface treatments. Systems within the same cluster tend to exhibit comparable error levels and wavelength trends, whereas isolated systems form distinct groups at large distances from the rest. Small distances indicate similar discrepancy patterns within the analysed dataset, separately from matched metrological behaviour, shared traceability, or a common calibration strategy. As a representative example, Figure 6 shows the technical distance matrix for R_a on Ti6Al4V after finish milling. Equivalent matrices for other material–process combinations and parameters are available in the supplementary materials, and they provide a compact complement to the decision tables (Tables 3–7) by showing which systems behave similarly *within a given material–process context*.

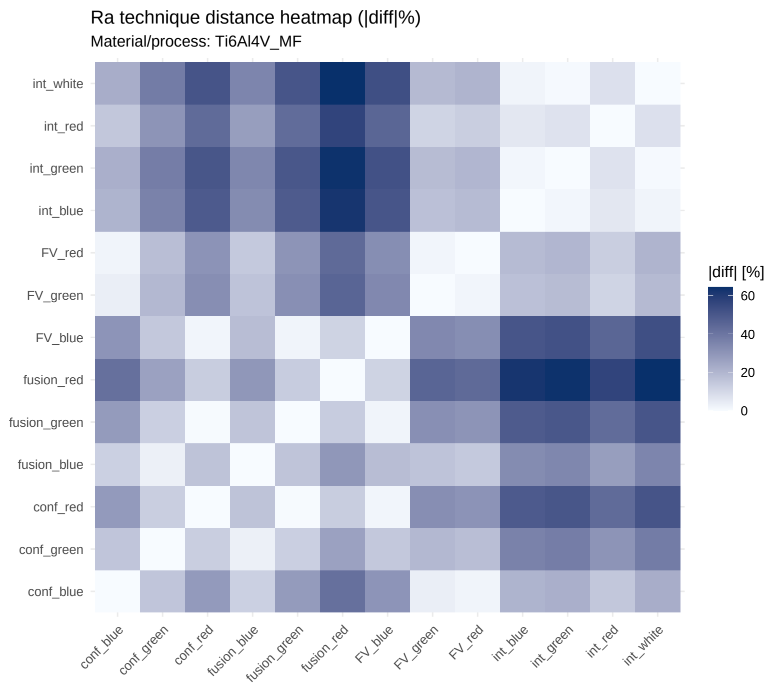

Figure 6: Technical distance matrix between optical and tactile systems for R_a on Ti6Al4V after finish milling, based on mean absolute roughness error $|\text{diff}|(\%)$

4 Discussion

A multi-material retrospective benchmark of optical workflows against an internal tactile baseline has been presented. Within the archive and its parameter-label comparison frame, three patterns are robust. First, system choice dominates the discrepancy structure across the archive. Second, illumination band matters, but mainly as a secondary, system-specific tuning variable. Third, performance is strongly parameter-dependent: amplitude-type parameters can enter a lower-discrepancy regime under selected configurations, whereas R_{sm} remains difficult.

4.1 Justification of discrepancy bands

The 10% and 30% discrepancy bands used in the decision summaries are operational triage bands, not tolerance limits, conformance criteria, or specification thresholds. The 10% band was chosen because it approximates the level at which optical–tactile discrepancy approaches the same order as typical production profilometer repeatability on engineering surfaces; below 10%, the optical workflow is a plausible screening candidate for workflow-specific prospective validation. The 30% band was selected because discrepancies above this level are large enough that tactile confirmation or a revised optical protocol is almost always the more defensible route, regardless of the specific material or application. The 10–30% middle band identifies cases where material–process-specific checking is advisable. These bands are descriptive screening heuristics, not formal acceptance criteria. Whether a particular discrepancy level is acceptable depends on the application tolerance, the functional role of the parameter, and the combined uncertainty budget—none of which can be prescribed by a retrospective archive-level benchmark.

4.2 Verification of unit and scaling consistency across parameters

The R_{sm} scale-sensitivity analysis (Table 5) raised the question of whether similar unit or scaling artefacts could affect other parameters. To check this, the retained optical and tactile outputs for the remaining strictly positive parameters (R_a, R_q, R_t, R_v, R_z, R_{z1max}) were examined for systematic order-of-magnitude offsets. For these amplitude parameters, optical and tactile values shared the same micrometre units in the retained instrument exports, and the median optical/tactile ratio ranged from 0.82 to 1.24 across parameters and systems—consistent with instrument-level differences in height response rather than unit-convention mismatches. The R_{sm} anomaly is specific to the spacing-parameter class: tactile R_{sm} was exported in micrometres while some optical R_{sm} exports were in millimetres, creating a factor of approximately 1000. No comparable unit mismatch was found for any amplitude parameter. The practical meaning of the x1000 sensitivity analysis is that exported R_{sm} values should be harmonised for unit convention before any comparison is attempted; the manuscript recommends that future raw-profile studies document the export unit for every spacing parameter explicitly. The x1000 analysis is a post hoc diagnostic illustration, not a corrected endpoint.

4.3 Impact of incomplete optical metadata and missing repeats

Important optical measurement settings—objective magnification, numerical aperture, lateral sampling interval, form-removal operator, filter cut-off, and profile-extraction rules—were not retained consistently across all instruments. This limitation means that the reported workflow-level discrepancies confound genuine instrument response differences with differences in post-processing settings chosen by individual operators. The present benchmark therefore characterises real-world exported-workflow behaviour, including operator- and software-specific processing choices, rather than isolating instrument physics. A prospective study that controls these settings would be needed to separate the instrument contribution from the processing-chain contribution. The retained metadata catalogue (Table 2) makes this limitation transparent.

Optical repeat measurements were also retained less consistently than tactile repeats. For most system–illumination entries, only one exported optical result was available per specimen. This constrains the reliability of optical-workflow ranking within individual parameter–surface groups, because the observed discrepancy for a given workflow may be influenced by single-realisation noise. Two mitigations are present. First, the fixed-workflow and transfer analyses aggregate across many groups, so ranking based on a single per-group optical value is averaged over dozens of independent groups. Second, the bootstrap intervals on aggregate medians (Supplementary Table S5) reflect the group-level resampling uncertainty. However, per-group ranking reliability cannot be fully assessed without optical repeats. A targeted repeatability study measuring each optical workflow 5–10 times on a small set of representative surfaces would be a valuable complement to the present archive-level screening and is recommended as part of the prospective validation phase.

4.4 Best- and worst-case material–process combinations

Table 7 and Table 6 identify the material and process strata with the most and least favourable optical–tactile agreement. Among materials, graphite (ELLOR) and MO58A brass generally exhibit the lowest retrospective discrepancies across amplitude parameters, consistent with their diffuse, non-specular surfaces that reduce speckle and coherence artefacts. Al7075 occupies an intermediate position. Ti6Al4V and 14301 stainless steel are systematically more challenging. For Ti6Al4V, the combination of relatively high reflectance in the visible range, sensitivity to surface oxidation and contamination, and the anisotropic, directionally textured topography produced by milling contribute to larger and more variable optical–tactile discrepancies. For 14301 stainless steel, the dominant factor is process-dependent: the rough face-turning and rough milling conditions produce highly irregular, reflective surfaces with deep tool marks that challenge the height-reconstruction range and speckle-handling capability of several optical systems. These physical attributions are qualitative and based on known optical behaviour of these material classes; a quantitative correlation with measured reflectance spectra or profilometric surface slopes would require dedicated optical-characterisation measurements that were not part of the archived dataset.

Among processes, honing, glass bead blasting, and wire EDM finishes generally enable the lowest discrepancies for amplitude parameters, likely because the isotropic or stochastically textured surfaces they produce reduce directionality artefacts and present favourable scattering characteristics to the optical instruments. Rough face turning and rough milling are the most challenging processes, producing deep, directional tool marks with steep local slopes and variable reflectance. The burnishing process occupies an intermediate position; while it produces smoother surfaces, the directional burnishing marks and altered near-surface reflectivity can still challenge some optical configurations.

To illustrate the contrast, a representative worst-case example is 14301 stainless steel after rough face turning (detailed in Section 3.3): for R_a, all optical workflows underestimate the roughness relative to the tactile baseline by 42–60%, and no system–illumination combination enters the $\leq 10\%$ band. The contrast with the Ti6Al4V finish-milling case (Section 3.2) illustrates the strong material–process modulation that underlies the aggregate summaries in Tables 7–6 and reinforces the central message that workflow selection is a per-group decision, not a universal recommendation.

In practice, practitioners can identify surfaces similar to those studied here through the material and process descriptors retained in the archive (Table 2). The process labels correspond to common, standardised manufacturing operations. A laboratory working with, for example, milled Ti6Al4V or turned 14301 stainless steel can consult the relevant decision-table stratum to assess whether optical screening is plausible or whether tactile confirmation should remain the default. The decision tables are most informative when the user's material–process combination closely matches one represented in the archive; for substantially different materials or processes, the benchmark provides a methodological template rather than directly transferable discrepancy values.

4.5 Workflow transferability by parameter, material, and process

The leave-one-surface workflow-transfer analysis (Table 9) reveals that transferability differs systematically across parameters. Amplitude parameters with the lowest retrospective discrepancies (R_t, R_v, R_z, R_{z1max}) also show the largest transfer penalty: their retrospective medians are 4.8–6.6%, but held-out transfer medians rise to 18–39%. This occurs because the low retrospective discrepancies for these parameters depend on precise workflow-to-surface matching, and the optimal workflow for one material–process group is often not the optimal workflow for another. R_a and R_q show more moderate transfer behaviour: their retrospective and held-out medians are closer, reflecting the greater robustness of these averaging parameters to workflow variation. Material-wise, graphite shows the best transferability (held-out median 15%), whereas MO58A brass, Ti6Al4V, and 14301 stainless steel occupy an intermediate range (held-out medians 30–36%), and Al7075 shows the poorest transferability (held-out median 52%). Process-wise, wire EDM and glass bead blasting show the best transferability across parameters, while rough face turning and rough milling show the poorest.

4.6 Practical guidance for system selection

The decision tables show that no single optical system dominates all parameters. Interferometry-based configurations are most frequently the lowest-discrepancy choice for amplitude parameters, but focus variation and confocal microscopy appear competitively in specific material–process strata. When a practitioner must select one system for multiple parameters, the fixed-workflow summaries in Table 8 provide the relevant guidance: interferometry with green illumination achieves a median discrepancy of 24.8% with complete coverage, but still leaves 77% of groups above 10%. If the application prioritises a specific parameter (e.g., R_a for general quality control), Table 3 identifies the system–illumination combinations most frequently associated with low discrepancy for that parameter. If the application involves multiple parameters, the practitioner should either (a) accept the higher fixed-workflow discrepancy levels reported in Table 8, (b) validate separate workflows for different parameter families, or (c) restrict the material–process range to strata where the chosen system performs well across all parameters of interest. The technical distance maps (e.g., Figure 6) can help identify systems that behave similarly, allowing substitution within a cluster without large changes in discrepancy.

4.7 Interpretation of R_{sm} and evidence for the spacing-parameter hypothesis

The interpretation that exported R_{sm} is dominated by unit conventions and lateral-sampling differences rather than by genuine surface-periodicity disagreement is supported by several lines of indirect evidence, although it cannot be confirmed without raw-profile data. First, the x1000 sensitivity analysis reduces the median discrepancy from 99.9% to 17.3% for the lowest-discrepancy workflows, which is consistent with a dominant unit-convention effect. Second, all four optical-system families show uniformly high R_{sm} discrepancy (Table 4), which is difficult to explain by instrument-specific physics alone. Third, amplitude parameters exported by the same instruments on the same specimens do not show comparable order-of-magnitude discrepancies, ruling out a global instrument malfunction or data-corruption hypothesis. However, the residual discrepancies after x1000 scaling (15 of 59 groups still above 30%) suggest that additional factors—lateral sampling density, profile-feature-recognition differences between optical and tactile instruments, and filter interactions—also contribute. A definitive test would require a prospective study that: (i) records the raw profile data from both tactile and optical instruments, (ii) harmonises the lateral sampling interval and evaluation length, (iii) applies identical filtering and profile-specification operators, and (iv) documents the export-unit convention explicitly. The present archive identifies R_{sm} as the highest-priority parameter for such a future raw-profile and resolution-controlled study.

4.8 Relation to ISO measurement requirements

ISO 21920-2 and ISO 21920-3 define profile parameters and specification operators but do not prescribe acceptable discrepancy limits between measurement methods. ISO 3274 specifies nominal characteristics of contact stylus instruments, and ISO 16610-21 defines the Gaussian profile filter. The present benchmark operates within this ISO terminology but does not claim that any particular discrepancy level satisfies a specific ISO requirement. In practice, whether a given discrepancy level is acceptable depends on the ratio of the measurement uncertainty to the specification tolerance. For a typical machining tolerance of ± 10–25 µm on a roughness parameter, a measurement discrepancy of 24.8% (the fixed-workflow median) may be acceptable if the absolute roughness value is small (e.g., $R_a = 0.4$ µm, discrepancy ≈ 0.1 µm) but becomes problematic if the roughness is larger (e.g., $R_a = 6$ µm, discrepancy ≈ 1.5 µm). The benchmark decision bands should therefore be interpreted relative to the application tolerance, not as absolute acceptability thresholds. Laboratories are encouraged to combine the benchmark screening results with their own gauge capability studies and uncertainty budgets.

4.9 Treatment of R_{sk} and alternative normalisation

Percentage-based R_{sk} discrepancy, defined analogously to Eq. 1, is intrinsically unstable because R_{sk} crosses zero for many engineering surfaces, producing near-zero denominators and inflated percentage values. The manuscript excludes R_{sk} from the main decision tables for this reason. The per-group percentage-based R_{sk} screening summaries are retained in the supplementary CSV output `results_decision_rsk_screening_groups.csv` for archive completeness. Because of the intrinsic instability of percentage normalisation for near-zero denominators, R_{sk} comparisons are recommended as supplementary screening descriptors only, not as primary workflow-selection criteria.

4.10 Data dictionary and independent verification

The public reproducibility package includes a data dictionary (`README.md` in the repository root) that documents the structure, variable names, units, and coding of every derived CSV output file used in the manuscript analyses. The analysis workflow is scripted end-to-end: R and Python scripts read the derived CSV inputs, perform all computations (ANOVA, regression, bootstrap, decision-table construction, and figure generation), and write the outputs to the `outputs/` and `plots/` directories. No manual calculation or spreadsheet-based step intervenes between the derived input files and the reported tables and figures. Because the raw `.sur` instrument-export files are excluded from the public package for size and transfer reasons, independent verification without raw files relies on the following pathway: (i) the derived CSV inputs contain the exported roughness parameter values, specimen identifiers, and workflow labels that constitute the comparison space; (ii) all analyses operate on these CSVs; (iii) the scripts are version-controlled and the repository is archived with a fixed DOI; (iv) a researcher can download the repository, run the scripts in the documented R and Python environments, and reproduce every number in the manuscript tables and supplementary tables. Full replication including the raw profiles would require access to the original `.sur` files, which are available from the corresponding author subject to data-transfer constraints.

4.11 Requirements for a prospective validation study

The present benchmark identifies candidate optical workflows for follow-up validation but does not itself establish optical–tactile equivalence. A complete prospective validation study would require the following elements:

1. **Sample design.** A minimum of 5–10 specimens per material–process–parameter stratum, with specimens drawn from independent production batches to separate batch-to-batch from measurement variability.

2. **Measurement protocol.** Both tactile and optical measurements performed on the same specimen locations, with documented stylus geometry (tip radius, cone angle), nominal contact force, evaluation length, and traversing length per ISO 3274 and ISO 21920-3; documented optical objective, numerical aperture, field of view, lateral sampling interval, and illumination settings; and a common, documented filter chain (e.g., Gaussian filter per ISO 16610-21 with a fixed cut-off wavelength λ_c) applied to both tactile and optical data at the raw-profile level.

3. **Repeatability and reproducibility.** At least 5–10 repeated measurements per instrument per specimen, performed under repeatability conditions, plus a limited between-operator and between-day reproducibility assessment on a subset of specimens.

4. **Uncertainty budget.** A combined uncertainty model incorporating tactile calibration uncertainty, optical vertical and lateral calibration uncertainty, repeatability components for both workflows, form-removal and filter-choice sensitivity, temperature effects, and specimen inhomogeneity.

5. **Acceptance criteria.** Pre-defined discrepancy thresholds linked to the application tolerance, expressed either as maximum permissible error

(MPE) relative to the tactile reference or as an E_n score incorporating the combined uncertainty of both workflows.

6. **Blinded or pre-registered design.** The optical workflow(s) to be validated should be fixed before data collection, not selected post hoc from the same dataset used for validation, to avoid the selection penalty quantified in the present benchmark.

The decision tables and fixed-workflow summaries in this manuscript serve as screening evidence to prioritise which material–process–parameter–workflow combinations merit the investment of such a prospective study.

4.12 Interpretive scope

Several features define the scope of the present conclusions. The study compares exported roughness parameters in a harmonised comparison space defined by matched material–process identifiers and same-named parameter labels. The reported discrepancies characterise workflow-level discrepancy within this comparison space. The tactile mean is an internal baseline derived from $n = 20$ repeats with an available repeatability descriptor (s_S, u_S). The tactile repeatability analysis indicates that the random variability of tactile measurements is usually smaller than the observed optical–tactile discrepancies, while unmodelled tactile bias or calibration effects may still shift individual discrepancies. Optical repeat measurements were retained less consistently than tactile repeats, so the optical side of the benchmark is an archived-output comparison whose per-group ranking reliability is limited; aggregate summaries are more robust (Section 4.3). The decision tables are retrospective, with the lowest-discrepancy system+colour selected after observing the data; the fixed-configuration and leave-one-surface transfer analyses quantify this selection effect from complementary perspectives. Bootstrap intervals in Supplementary Table S5 describe stability of the archive-level median summaries, not formal uncertainty of future optical use. Wavelength regressions and the descriptive summaries in Supplementary Tables S2–S3 are often based on only three or four colour levels per system and provide screening trends. Raw profiles, common filtering settings, complete optical instrument metadata, detailed process settings, and a combined tactile–optical uncertainty model were retained unevenly across the archive. The R_{sm} scale sensitivity and R_{sk} percentage-based screening are post hoc diagnostics, not primary results.

5 Conclusions

Within this scope, the main findings of the multi-material benchmark can be summarised as follows:

- **System choice is the primary workflow driver.** Across the ANOVA screens, the measurement system has a much larger effect size than illumination colour in most groups (median $\eta^2_{\text{system}} = 0.651$ versus $\eta^2_{\text{colour}} = 0.106$), while wavelength/colour behaviour remains a system-specific tuning issue rather than a universal correction (Supplementary Tables S1–S2).
- **Parameter dependence is substantial.** Retrospective lowest-discrepancy medians differ sharply across the strictly positive parameters: amplitude-type measures such as R_t, R_v, R_z, and R_{z1max} achieve medians of 4.8–6.6%, R_a and R_q remain below 10%, and retained exported-value R_{sm} remains a harmonisation-sensitive spacing diagnostic (Tables 3–5).
- **Fixed and transferred workflow choices carry a clear penalty.** The lowest-median complete-coverage fixed workflow has median discrepancy 24.8%, while leave-one-surface workflow transfer yields a held-out median of 32.8%, with 52% of held-out groups above 30% (Table 8, Table 9, and Figure 1).
- **Tactile Type A uncertainty is low relative to most optical discrepancies.** Across all 472 parameter–surface groups, the tactile baseline is based on $n = 20$ repeats. For the seven strictly positive parameters, the median relative Type A standard uncertainty of the tactile mean is 1.45% (Q1–Q3: 0.85–2.44%), supporting its use as an internal repeatability benchmark, although not replacing a full uncertainty budget.
- **The main output is a benchmark for follow-up selection.** The decision summaries identify candidate optical workflows for prospective validation in specific parameter–surface groups. They prioritise follow-up experiments rather than establishing a universal fixed optical setup or formal tactile/optical equivalence.

Overall, the benchmark identifies where optical roughness workflows are plausible candidates for prospective validation and where tactile confirmation remains the more defensible route. Its main value is not an equivalence claim, but a reproducible decision map separating local low-discrepancy matches from fixed-workflow and held-out transfer performance.

Supplementary materials

The following supporting information is available online at the journal website:

- **Supplementary S1**: additional tables and figures extending the Results. The material includes (i) two-way ANOVA main effects and η^2 effect sizes for the factors system and colour (Supplementary Table S1), (ii) system-wise slopes of $|\text{diff}|(\%)$ versus illumination wavelength (Supplementary Table S2), (iii) descriptive statistics by system (Supplementary Table S3), (iv) parameter-wise internal tactile repeatability summaries (Supplementary Table S4), and (v) bootstrap percentile intervals for aggregate retrospective lowest-discrepancy medians (Supplementary Table S5), together with representative plots for selected material–process combinations. The supplementary tables are generated from the same archived output files as the manuscript tables, and the generation script is included in the public repository.

- **Data dictionary**: a file `README.md` in the repository root documents the structure, variable names, units, and coding of every derived CSV output file, together with instructions for reproducing the analyses from the archived repository.

- R_{sk} **screening**: the file `results_decision_rsk_screening_groups.csv` in the repository `outputs/` directory contains the per-group percentage-based R_{sk} screening summaries for archive completeness. Because of near-zero tactile denominators, these percentage-based values are retained as secondary descriptors only and are excluded from the main manuscript decision tables.

CRediT authorship contribution statement

The author-declared contribution shares are as follows: (DK), 41%; (PN), 14%; (JK), 8%; (GK), 8%; (KN), 7%; (NW), 7%; (ŁŚ), 7%; (MW), 4%; and (BG), 4%.
(DK): Conceptualisation, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing. **(PN):** Conceptualisation, Supervision, Project administration. **(JK):** Supervision, Resources. **(GK):** Supervision. **(KN):** Investigation, Validation.

(NW): Investigation, Data curation. **(ŁŚ):** Validation, Formal analysis. **(MW):** Supervision. **(BG):** Supervision.

Conflicts of interest

The authors declare no conflict of interest.

Data availability

The R/Python analysis scripts, configuration files, manuscript and supplement sources, generated CSV outputs, generated tables, and generated figures supporting the findings are provided as submission reproducibility materials. Public repository and archive identifiers are supplied separately for editorial handling and can be inserted in the final version according to journal anonymisation policy. Raw `.sur` surface files are excluded from the public package as original instrument exports with size and transfer constraints; the derived CSV outputs needed to reproduce the manuscript tables, fixed-workflow sensitivity analysis, workflow-transfer check, global heatmap, selection-penalty figure, and supplementary tables are included. The excluded raw `.sur` files are available from the authors on reasonable request, subject to data-transfer constraints.

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